

Propellantless electron thruster for LEO station keeping

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Abstract

Chip-scale spacecraft lose their orbit to atmospheric drag fastest of all: a 3 g body 10 mm across and 30 mm long re-enters from 400 km in 41–46 days and from 550 km within a year near solar maximum, and no flown propulsion system is small enough to carry. This paper describes a thruster with no tank, no feed system and no neutralizer. A cathode held a few hundred volts below the spacecraft body emits electrons through a single aperture. The escaping beam produces thrust and leaves the body positively charged, so the body collects electrons from the ionospheric plasma until collection equals emission. The circuit closes through the ionosphere with zero net mass flow.

Particle-in-cell simulations (WarpX) of a 10 mm conducting body in a dayside ionospheric plasma measure the complete operating envelope over 100–350 V: 3.4 to 43 nN of thrust from 12 to 278 mW of beam power, with 96–99 % of the beam escaping, a floating potential of 4–48 V, and an energy efficiency of 0.68–0.77. Two measured constants describe every run to within 3 %. Year-long orbit propagations at 400–700 km on near-equatorial, ISS and sun-synchronous orbits, near solar maximum (2024) and minimum (2019), with the supply voltage following the drag and the floating potential solved from the plasma along the orbit, show the 3 g body holding 500–700 km for a mean beam power of 2.5–48 mW at solar maximum and 0.1–2 mW at minimum, inside the tested 100–350 V envelope at least 97 % of the time and never more than 10 m below its target altitude; at 400 km it needs 21 mW at solar minimum, while near maximum the demand exceeds the simulated envelope. Because both drag and electron collection scale with area, the feasibility condition depends on shape, not size: a 3U CubeSat of the same shape needs about 1 W at 600 km near solar maximum and tens of milliwatts near minimum, an estimate in a sheath regime not yet simulated. A bench experiment confirms the electrode topology on hardware. A three-dimensional follow-up campaign puts the geomagnetic field perpendicular to the beam, the orientation the mission flies in: at flight strength the operating point survives (thrust within 0.8 %, escape unchanged, a 2.6 V float shift at the settled equilibrium), while at ten times flight strength the return circuit cannot close and the device must fire along the field. Open items are the choice of flight emitter and the CubeSat sheath regime.

1 Introduction

Small spacecraft below 600 km lose their orbits to atmospheric drag within months to a few years, and the smallest lose them fastest: drag grows with area and inertia with volume. Near solar maximum a 3 g chip-scale body, 10 mm across and 30 mm long, re-enters from 400 km in 41–46 days and from 550 km in 300–329 days, while a 4 kg body of the same shape at CubeSat size lasts about a year from 400 km (Sec. 5). Most carry no propulsion at all.

The thrust to hold altitude is nanonewtons to a few micronewtons, well within the range of flight-proven micropropulsion. Colloid thrusters on LISA Pathfinder resolved 0.1 μN [1]; NanoFEEP flew on a 1U CubeSat at 0.2–0.3 W [2]. Every one of these systems spends a stored propellant through tanks, valves, feed lines or precision emitter heads. The propellant imposes a total-

impulse limit, and the hardware adds mass and cost that are rarely justified on a CubeSat.

This paper describes a thruster with none of that hardware. A cathode held below the spacecraft potential emits electrons through an aperture. The escaping beam produces thrust and leaves the body positively charged, so the body collects ambient electrons from the ionospheric plasma until collection balances emission. The circuit closes through the ionosphere with no propellant and no net mass flow. An electron beam carries about 200 times less momentum per watt than an ion beam, but LEO drag on a small spacecraft is nanonewtons, and at that scale the beam costs tens of milliwatts. The whole device is a cathode, an aperture and a high-voltage supply. It carries no propellant, no tank and no feed system. On a chip-scale spacecraft, where nothing flown fits, it holds 500–700 km on milliwatts that the body's own solar cells can supply; the same device on a CubeSat needs watts.

The following sections describe the operating principle, test each claim through a ladder of particle-in-cell simulations that build from a vacuum electron gun to the full device in ionospheric plasma, and assess the result against a year of real drag at 400–700 km on three orbit families, at both ends of the solar cycle, and against the lifetime of the same spacecraft without the thruster.

2 Concept

The device has three parts (Fig. 1):

1. a cathode, held at $-V$ with respect to the spacecraft body, that emits electrons and accelerates them out through an aperture;
2. the spacecraft's own conducting skin, which is the second electrode and collects electrons from the plasma;
3. a high-voltage supply between the two, which moves collected electrons back to the cathode.

Once switched on, the device reaches a steady state:

1. the spacecraft floats in the ionospheric plasma;
2. the supply is switched on and the cathode emits;
3. each escaping electron leaves the body one charge more positive;
4. the positive body attracts electrons from the plasma;
5. collection grows until it equals emission. The body settles at a potential φ where the two currents balance.

Two questions decide whether this works.

Why does the positive body not recapture its own beam? Because the beam electrons carry 100–300 eV while φ is a few volts to a few tens of volts; ambient electrons at 0.1 eV are captured, beam electrons are not.

Does the returning current cancel the thrust? The collected electrons gain 4–48 eV from the floating potential, against 77–272 eV for the beam, and they arrive from all directions rather than straight astern. The simulations measure the net cancellation at 0.4–1.4 % of the beam thrust at 200–350 V, rising to 9 % at 100 V where the slower beam carries less momentum relative to the collected electrons (Sec. 4.4.1).

3 Theory

3.1 Thrust

The beam is a current I of electrons with kinetic energy KE. Its momentum flux is the thrust:

$$F = \frac{I\sqrt{2m_e \text{KE}}}{e} = 3.37 \frac{\text{nN}}{\text{mA}\sqrt{\text{eV}}} I\sqrt{\text{KE}}. \quad (1)$$

The constant contains only the electron mass and charge. The exhaust velocity is $v_e = \sqrt{2e \text{KE}/m_e}$, of order 10^7 m/s.

3.2 The float tax

The body floats at $+\varphi$ and the cathode at $\varphi - V$, so the accelerating potential is $V - \varphi$. Space charge near the cathode depresses it further. The exhaust energy is

$$\text{KE} = \kappa (V - \varphi), \quad (2)$$

where κ is the fraction of the drop that reaches the exhaust. In a steady field an electron emitted at cathode potential leaves with $e(V - \varphi)$, so a real cathode has $\kappa \approx 1$; the simulations measure $\kappa \approx 0.81$ because their beam is launched just above the cathode (Sec. 4.4.1). The ratio φ/V is the fraction of supply voltage spent on collecting the return current.

3.3 Power and efficiency

Beam power is $P = VI$. Jet power over supply power is

$$\eta = \kappa \frac{V - \varphi}{V} f_{\text{esc}}, \quad (3)$$

with f_{esc} the fraction of the beam that clears the body. Eliminating I between Eq. (1) and $P = VI$ at fixed thrust gives $P \propto F\sqrt{V}$: doubling the voltage costs about 40 % more power for the same thrust. Thrust per watt is $F/P \approx 3.37\sqrt{\kappa}/\sqrt{V}$ in $\mu\text{N/W}$ with V in volts, about $0.2 \mu\text{N/W}$ at 200 V.

3.4 Emission ceiling

Low voltage is cheap, but the current a gun can source is limited by space charge. The planar Child–Langmuir scale for the simulated spot ($r = 0.5$ mm, gap 4.7 mm) is $I_{\text{CL}} \propto V^{3/2}/d^2$, equal to 0.08, 0.24 and 0.43 mA at 100, 200 and 300 V; the real gun emits 1.46 times that. Maximum thrust therefore scales as $I_{\text{CL}}\sqrt{V} \propto V^2$, and at 100 V the spot sources about 0.12 mA whatever the plasma does. In the $\varphi \ll V$ limit the best operating point is the lowest voltage whose emission ceiling still meets the thrust demand; with the float included, the minimum-power voltage sits somewhat higher (Sec. 5). A shorter gap or a wider spot moves the ceiling without changing the voltage.

3.5 Current collection

The body collects the one-sided thermal electron flux [6]

$$I_{\text{th}} = e A n_e \sqrt{\frac{kT_e}{2\pi m_e}}, \quad (4)$$

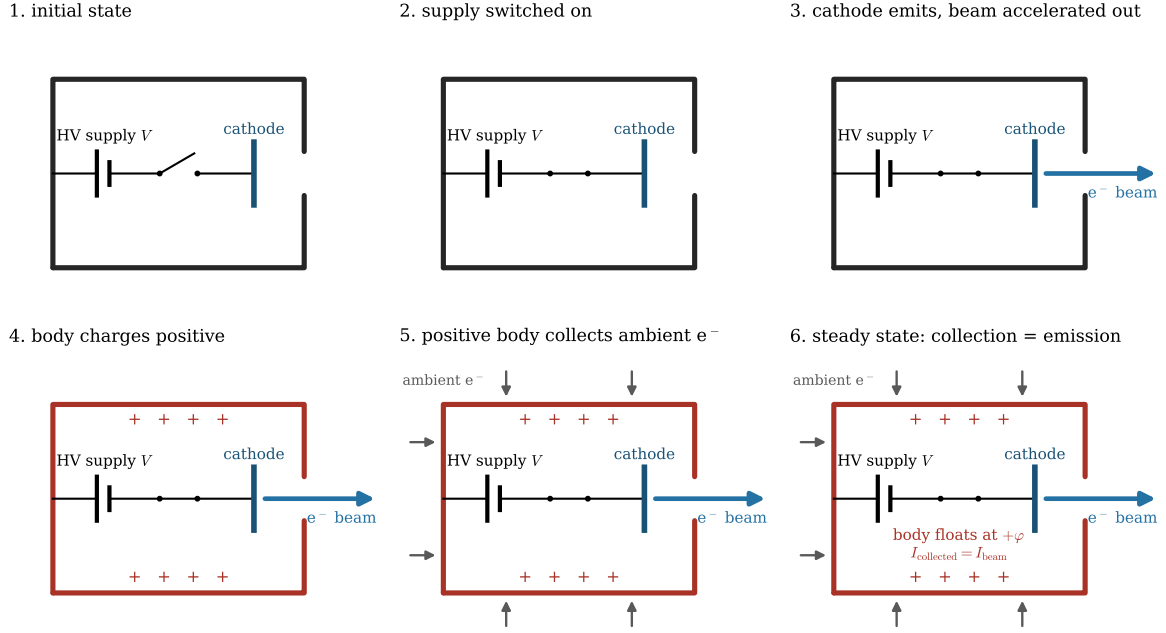


Figure 1: Operating sequence. The supply is switched on, the cathode emits, the escaping beam charges the body positive, and the positive body collects ambient electrons until collection equals emission.

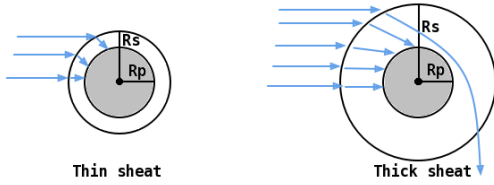


Figure 2: Thin sheath ($r_p \gg \lambda_D$): electrons reach the body on nearly straight paths and collection scales with surface area. Thick sheath ($r_p \lesssim \lambda_D$): the potential extends far beyond the body and bends electron orbits inward, giving the $(1 + \chi)^\alpha$ enhancement of Eq. (5). The 10 mm body sits in the thick-sheath regime; a CubeSat would sit in the thin-sheath regime.

enhanced by the positive potential. With $\chi = e\varphi/kT_e$,

$$I = \beta I_{th} (1 + \chi)^\alpha, \quad (5)$$

where $\alpha = 1$ for a sphere and 0.5 for a long cylinder in orbital-motion-limited theory, and β is a geometry factor. Setting I equal to the escaping current and inverting gives the float. Its stiffness goes as $1/(n_e \sqrt{T_e})$, so thin night-side plasma makes the body float higher for the same current.

3.6 Scale

Drag is proportional to the drag area S_{ref} : the ram face plus free-molecular friction on the walls parallel to the flow, whose coefficient is about 3 % of the ram one per unit area but which adds 37 % on an end-on body with $L/r = 6$. Collection and body-mounted power are proportional to the skin area. Thrust demand, beam current and power all scale with S_{ref} , so power per unit drag area depends only on altitude and voltage. Vehicle size drops out and what remains is the shape ratio A_{skin}/S_{ref} . A 10 mm can and a 3U CubeSat with the same shape have

the same feasibility margin. Mass does not enter the demand, since holding altitude takes thrust equal to the drag force; it sets only how fast a craft sinks while thrust falls short, and at equal density a 3U sinks ten times more slowly than the 10 mm can for the same shortfall.

3.7 Control

I and φ are plain electrical measurements, so Eqs. (1)–(2) let the spacecraft compute its own thrust in flight. Density, temperature and day/night all collapse into where the body floats, so a two-line servo on the measured float is the whole control law. A microcontroller is sufficient.

4 PIC simulations

4.1 Setup

The ladder and characterization runs use WarpX [3] (built on AMReX [4]) in RZ electrostatic mode with the parameters of Table 1; the transverse-field campaign of Sec. 4.4 uses the same code in Cartesian 3D. The body is a closed can with a cathode disk inside and an aperture in the lid, modelled as an embedded conductor (Fig. 3). Its potential is not prescribed. Each step the code counts the charge that left through the aperture and the charge that landed on the skin, and moves the body potential by the net charge over the body's self-capacitance, which is measured once from Gauss's law on a vacuum Laplace solve. The float is therefore an output, and ladder stage 7 checks the method on a bare body against the theoretical floating potential. Ions are repelled by the positive body, so the balance is between the escaping beam and collected electrons only; the reduced ion mass affects how

Table 1: PIC configuration of the full thruster at 200 V. Characterization runs change one entry at a time (Table 4).

	parameter	value
geometry	body outer diameter $\times$ length	10×5.5 mm
	wall thickness	0.4 mm
	exhaust aperture diameter	4 mm
	cathode disk diameter	3 mm
	emission spot diameter	1 mm
	cathode–lid gap	4.7 mm
electrical	supply voltage V	200 V
	beam current I (prescribed)	0.342 mA
	gun loading I/I_{CL}	1.46
	body self-capacitance	0.5–1 pF
plasma	n_e	$1.63 \times 10^{12} \text{ m}^{-3}$
	kT_e	114 meV
	kT_i	81 meV
	ion mass	$400 m_e$
numerics	cell size $\Delta r = \Delta z$	0.15 mm
	domain $r_{\max}$	30 mm
	particles per cell per species	16
	run length	800 ns (29 RC times)
	supply / gun switch-on	100 / 150 ns

fast the sheath settles, not where it settles. The ambient plasma is a dayside 400 km row kept constant by a particle reservoir. Every run freezes its configuration and passes fixed acceptance gates on escape fraction, floating potential, current balance and momentum before it is used.

4.2 Validation ladder

A PIC simulation of the full device has too many parameters to validate at once. The ladder isolates each piece of physics and checks it against theory before combining them, so the full thruster result inherits confidence from every stage below it. Nine stages build from a vacuum electron gun to the full device (Table 2). All nine pass.

Stages 1–3 test the gun alone in vacuum. Stage 1 confirms that a negative cathode emits and accelerates electrons through a near-neutral aperture, the basic electrode topology, with a $35 \mu\text{V}$ energy error on 100 V; the same topology is confirmed on hardware in Sec. 7. Stage 2 measures aperture transmission at three hole sizes (transmitted fractions 0.97, 0.90 and 1.00, matching the predicted values). Stage 3 confirms transmission is independent of voltage (0.006 pp spread over 200–300 V). These are current fractions; the energy fraction κ of Sec. 4.4.1 has a different origin.

Stages 4–7 test collection on a conducting body in the plasma. Stage 4 reproduces the thermal electron flux of Eq. (4) to within 1 %. Stages 5–6 bias the body to +3 and +10 V and measure 0.85 and 0.81 of the Laframboise OML ceiling, with the sheath growing from 4.1 to 6.9 mm as the potential rises. Stage 7 lets the body float freely and measures -0.251 V, inside the bracket between two

Table 2: Validation ladder.

#	stage	validates
1	negative cathode	emits through neutral aperture
2	holed anode	aperture controls transmission
3	voltage bracket	transmission is V -independent
4	thermal collection	PIC current vs Eq. (4)
5	biased, +3 V	OML collection at bias
6	biased, +10 V	larger sheath, more current
7	floating body	unbiased float vs theory
8	two-node Laplace	two potentials, one body
9	full thruster, 200 V	gun + plasma combined

Table 3: Steady-state values of the full thruster at 200 V. This run is the anchor for all characterization.

quantity	value
floating potential φ	17.0 V
beam escape f_{esc}	98.4 %
thrust F	13.65 nN
beam power P	68.4 mW
exhaust energy KE	147.5 eV
energy efficiency η	0.73

theoretical models. Stage 8 places two prescribed potentials on one embedded conductor and recovers the exact Laplace solution with zero violation, which validates the multi-potential method used in stage 9. Stage 9 is the full thruster, described next.

4.3 Full thruster

Stage 9 puts the gun inside the floating can and switches it on in the 400 km plasma (Fig. 3). The body charges positive as beam leaves, collection rises with the float, and the two currents balance within a few RC times. Table 3 lists the steady state at 200 V. The body floats at 17 V, 98.4 % of the beam escapes, and the thrust is 13.65 nN at 68.4 mW of beam power. This run is the anchor: everything in the next section perturbs it one axis at a time.

4.4 Characterization

Eight runs vary the full thruster one axis at a time: voltage first, then shape, plasma density and magnetic field, and one run that combines two axes (Table 4). All pass their gates. Predictions were written down before each run.

4.4.1 Voltage envelope

Three runs at 100, 300 and 350 V hold the same fraction of the Child–Langmuir current and the same plasma (Table 5, Fig. 4). Pre-registered thrust predictions landed within 1 %.

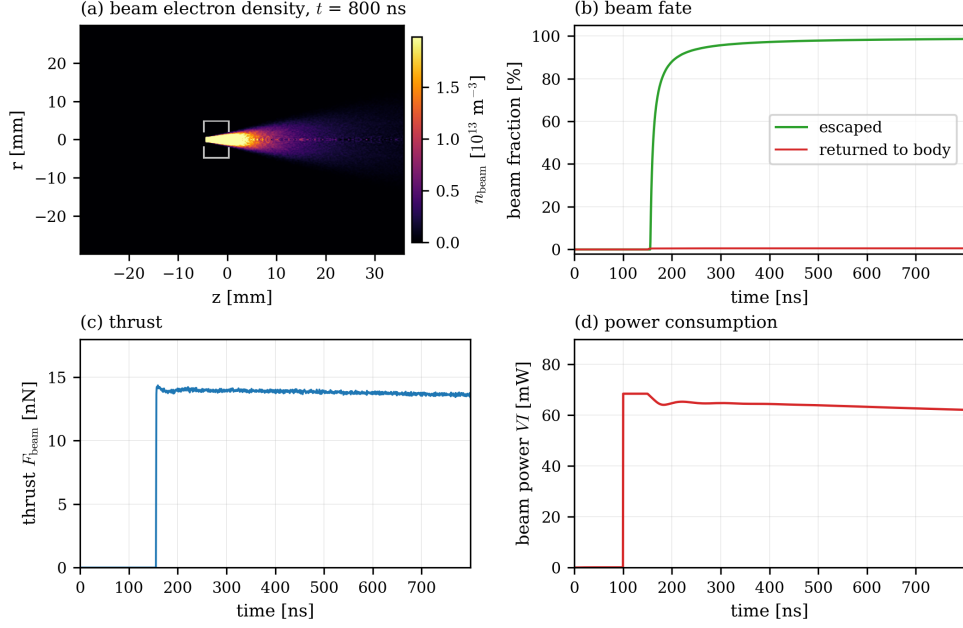


Figure 3: The full thruster at 200 V, the reference run for everything that follows. (a) Beam electron density at the end of the run; the grey outline is the can. (b) Fraction of emitted beam that has escaped the domain or returned to the body. (c) Beam thrust. (d) Drive power left after the float, $(V - \varphi) I$. The gun switches on at 150 ns; thrust, power and escape are all steady well before 800 ns.

Table 4: Characterization runs. Each changes one axis of the full thruster except the last, which combines two.

axis	φ [V]	F [nN]	f_{esc} [%]	KE [eV]
full thruster (200 V)	17.0	13.65	98.4	147.5
100 V	5.4	3.42	96.1	77.2
300 V	36.3	30.13	99.0	210.1
350 V	48.3	40.48	99.1	239.0
slender body ($L/r = 6$)	4.4	14.22	98.4	159.7
density $n_e/3$	42.5	12.39	99.1	122.0
$B_z = 30 \mu\text{T}$	17.2	13.64	98.4	147.3
$B_z = 300 \mu\text{T}$	48.6	12.06	98.3	115.9
350 V + slender	14.0	43.33	99.1	272.7

Table 5: Measured operating envelope of the 10 mm body. Grid-resolution uncertainty: $F \pm 4\%$, $\text{KE} \pm 7\%$, $\varphi \pm 2\%$.

V V	I mA	φ V	F nN	P mW	f_{esc} %	η
100	0.121	5.4	3.42	12.1	96.1	0.74
200	0.342	17.0	13.65	68.4	98.4	0.73
300	0.630	36.3	30.13	189	99.0	0.69
350	0.793	48.3	40.48	278	99.1	0.68

Two constants fit all four points:

$$F [\text{nN}] = 3.27 I [\text{mA}] \sqrt{\text{KE} [\text{eV}]}, \quad (6)$$

$$\text{KE} = 0.81 (V - \varphi). \quad (7)$$

The thrust slope is 97 % of Eq. (1); the deficit is plume divergence. $\kappa = 0.81$ is set by where the simulated beam is born, not by gun physics: it is launched 0.3 mm above the cathode, where a vacuum field solve puts the potential 19 % of V above the cathode, so every electron misses that part of the drop. The missing energy is 17–18 % of V in every run at this loading, independent of float, density and field, and it shrinks at higher loading as beam

space charge pulls the launch point down, the opposite of a space-charge depression. A real cathode emits at cathode potential and would leave the beam with $e(V - \varphi)$: about 11 % more thrust at the same current and η near 0.90. This paper keeps the measured κ , so its thrust and power figures are conservative by that margin. Thrust per watt follows $1/\sqrt{V}$ within 2.5 % over 100–300 V, and the 350 V point sits 3 % below it because the float tax has grown to 14 %. The energy efficiency falls slowly from 0.74 to 0.68 as the float tax grows; almost every electron escapes, and the rest of the gap to 1 is the launch deficit above.

The 300 V run also selects the collection exponent. Three values of α in Eq. (5) agree at 200 V and predict 31, 36 and 90 V for the 300 V float; the measurement is 36.3 V. A fit across the envelope gives $\alpha = 0.82$ – 0.89 , between the sphere and cylinder limits. The 350 V float was predicted from that fit at 47 V and measured at 48.3 V. One caveat applies to both high-voltage points: the sheath settles on the ion timescale, and their floats were still rising slowly at run end. The quoted values are tail averages; extrapolating the decaying slope puts the settled 300 V float at 42–48 V and the settled 350 V float past 50 V. No float value is a design limit; the float is a tax on the drive through Eq. (2), 14 % of the supply for the compact body at 350 V against 4 % for the slender body, which is why the slender body matters at 400 km.

4.4.2 Shape, density and field

Shape. Does elongation hurt current collection? Stretching the can to $L/r = 6$ multiplies the skin by 3.5 at the same ram area. Two hypotheses were recorded

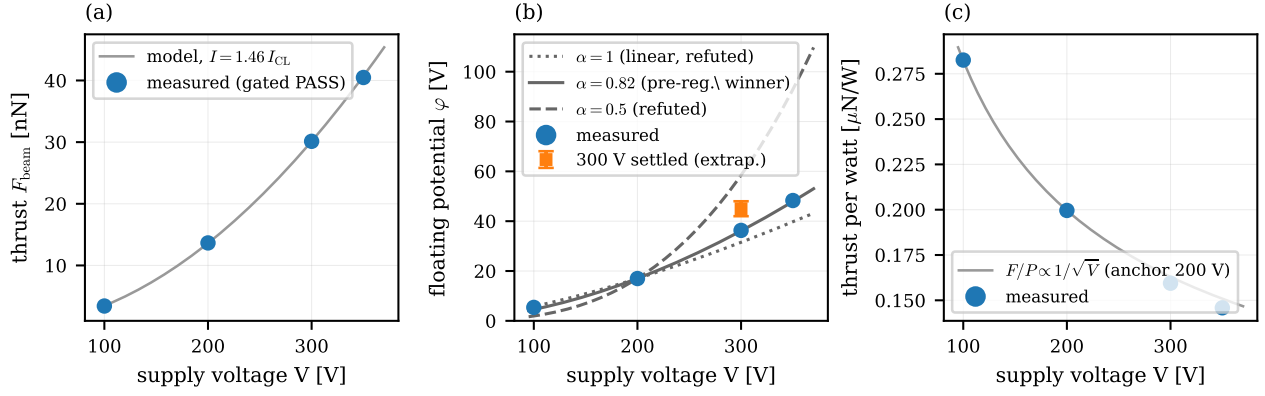


Figure 4: The four envelope runs (dots) against the model (lines). Left: measured thrust and the two-constant law $F = 3.27 I \sqrt{KE}$, $KE = 0.81(V - \phi)$. Centre: measured floating potential and the float predicted by Eq. (5) for three collection exponents, each forced through the 200 V point; they diverge above 200 V, and the 300 V measurement selects $\alpha \approx 0.82$ – 0.89 and refutes the sphere ($\alpha = 1$) and cylinder ($\alpha = 0.5$) limits. The square is the 300 V float extrapolated to sheath settlement. Right: measured thrust per watt and the $1/\sqrt{V}$ law that follows from $F \propto I\sqrt{V}$, $P = IV$, anchored at 200 V; the 350 V point falls below it because the float tax has grown.

before the run: area-only scaling predicted 4–5 V, and the long-cylinder limit predicted tens of volts. The float dropped to 4.4 V and the thrust rose 4 %, confirming area-only scaling. The pre-registered bracket used a $3.2\times$ area ratio in error; with the correct $3.5\times$ it is 3.8–4.3 V, 2 % below the measurement, so collection grows slightly slower than area. Fitted to both slender runs (200 and 350 V) at the shared exponent, the slender body collects like 3.1 times the compact body’s effective area; the mission model uses that measured value. A 3U CubeSat flying end-on has nearly the same shape ratio (Sec. 5.4).

Density. Does thin night-side plasma break the device? At one third density the run was extended to $2.4\mu\text{s}$ (three times the RC time) and the float settled at 42.5 V. Every fixed-exponent prediction from Eq. (5) overshoot, from 53 V at $\alpha = 1$ to 68 V at $\alpha = 0.82$. The fitted law overestimates the float in thin plasma, so night-side predictions from Sec. 5 are conservative.

Magnetic field. Does the geomagnetic field tax collection? Along the thrust axis, no. An axial field at flight strength ($30\mu\text{T}$) leaves float, thrust and escape within 0.2 V, 0.1 % and 0.06 percentage points of the reference run; at ten times flight strength collection is taxed, the float rises 33 V, and the thrust law absorbs the change once the measured float is inserted. The transverse orientation, the one the mission flies in, was then measured on a Cartesian 3D deck: the same body resolved at 1 mm cells in a $\pm 60\text{ mm}$ box, the escaped beam prescribed at the lid with the anchor’s measured energy, and the thrust ledger corrected for the Lorentz force on the particles in flight, since the exit flux alone underreads the emission reaction once the beam curls. Runs of $6\mu\text{s}$ settle the float on the model’s ion clock; the 800 ns snapshots reach about half the settled value, and thrust

moves under 4 %. At flight strength the operating point survives: thrust within 0.8 %, escape unchanged, settled float 29.4 V against the control’s 26.8 V, a 2.6 V tax, about 1 % of the drive. At ten times flight strength the circuit does not close: the float climbs through the 150 V abort ceiling with no equilibrium below 91 % of the beam energy, so in amplified fields the device must fire along the field, the mode the axial runs validate.

400 km operating point. Does the slender body at 350 V close the 400 km mean drag (32.9 nN)? The voltage and shape laws together predicted 11–17 V and 42–43 nN. It measured 14.0 V and 43.33 nN, covering the mean demand with a float well inside the 50 V limit. The two laws compose, which turns two one-dimensional curves into an operating map over 100–350 V and $L/r = 0.6$ – 6 .

5 Mission assessment

5.1 Method

Drag demand comes from TudatPy [5] propagation with NRLMSISE-00 density driven by measured F10.7 and A_p , and IRI-2020 plasma along the orbit. The test body is the slender can of the shape runs (10 mm diameter, 30 mm long, $L/r = 6$), flown axial: the thrust axis points along the velocity, which for $L > \frac{\pi}{2}r$ is also the low-drag pose. Its drag area S_{ref} is the ram cap plus free-molecular friction on the side wall, with $C_d = 2.2$. Its mass, 3.1 g, is the density of a 4 kg 3U; mass does not enter the thrust demand. Circular orbits at 400–700 km on three families (near-equatorial, $i = 0.5^\circ$; ISS, 51.6° ; sun-synchronous with a 10:30 ascending node) are flown for a year near solar maximum (2024) and minimum (2019). In station keeping the drag is cancelled exactly, so the drag at each 5-minute point is the thrust demand. In free

Table 6: Mean drag on the 3 g slender body and its lifetime with the thruster off. Ranges span the three orbit families; the 3U column is the same shape scaled $\times 10$ (4 kg).

altitude	drag [nN]		free fall, 3 g		3U 2024
	2024	2019	2024	2019	
400 km	38–42	3.8–4.8	41–46 d	171–241 d	1.0–1.1 yr
500 km	8.6–9.8	0.5–0.6	179–204 d	3.3–3.5 yr	> 5 yr
550 km	4.3–5.0	0.2	300–329 d	4.3–4.5 yr	> 5 yr
600 km	2.3–2.6	0.1	1.8–2.2 yr	> 5 yr	> 5 yr
650 km	1.2–1.4	0.05	> 5 yr	> 5 yr	> 5 yr
700 km	0.7–0.8	0.03	> 5 yr	> 5 yr	> 5 yr

fall the thruster is off and the orbit decays to re-entry (120 km) or a 5-year cap. The mission model uses the slender body’s own collection prefactor (Sec. 4.4). The PIC plasma row of Table 1 sits near the dayside top of the 400 km range; night-side and high-latitude densities are 5–40 times lower, which is where collection is weakest.

5.2 Station keeping against free fall

Without the thruster the 3 g body re-enters from 400 km in 41–46 days and from 550 km within a year near solar maximum (Table 6, Fig. 5a). With it, the voltage-following model holds 500–700 km all year for 2.5–48 mW of mean beam power at solar maximum and 0.1–2 mW at minimum, with at least 97.5 % of the orbit points inside the tested 100–350 V envelope (Table 7, Fig. 5b). Replaying each year with thrust capped at the measured capability at 350 V, firing at full capability whenever the craft is below its target and integrating any shortfall into an altitude deficit ($\dot{a} = 2aF/mv$), the orbit never drops more than 10 m below its target at 500–700 km. Solar activity moves the demand 15–25 times; inclination moves it about 10 %, partly because an inclined circular orbit flies 10–15 km higher on average over the oblate Earth. The high-latitude orbits see thinner plasma and float higher, which costs supply voltage at those points, not closure: no voltage or float value is a design limit in this model, and every point of both years has an operating point. At 400 km the solar-maximum peaks (119–128 nN) exceed what 350 V delivers even on average, so that case is not held inside the simulated envelope; at solar minimum 400 km is held for 21 mW. Two parts of Table 7 are extrapolations, flagged per point in the model output: the collection law below about a third of the simulated density, where the one off-density run found it conservative, and the calibration on 800 ns floats (Sec. 8).

5.3 Power

Power is the thruster’s only interface to the spacecraft. Body-mounted cells on the slender can supply about 37 mW orbit-averaged (30 % cells on a third of the skin, 25 % illumination; an example, not part of the thruster).

Replayed within that budget, the orbit holds at 550–700 km near solar maximum, with dips up to 1.2 km at 550 km, and at every altitude near solar minimum; at 500 km near maximum the 46–48 mW demand exceeds it. Supply and demand both grow with area, so the ratio holds with size.

5.4 Scaling to CubeSats

By the argument of Sec. 3, a 3U CubeSat flying end-on ($10 \times 10 \times 30$ cm) has nearly the shape ratio of the slender can and the same feasibility margin: against body-mounted cells it closes 1.1, 2.2 and 4.2 times over at 500, 550 and 600 km near solar maximum. In absolute terms it needs about 4–5 W at 500 km, 2 W at 550 km, 1 W at 600 km and 0.3 W at 700 km near solar maximum, and 0.2 W, 80 mW, 40 mW and 13 mW near minimum. That puts a 3U in the watt class where electrospray and FEEP already fly (Sec. 6), and above 500 km its free-fall lifetime already exceeds five years (Table 6); for a CubeSat the device is a propellantless option for altitude control, while for a chip-scale body it decides whether the mission survives. Mass does not change these powers: a 4 kg 3U sinks about 50 m/day at 500 km with the thruster off, against 0.5 km/day for the 3 g can, so it tolerates shortfalls far better. Larger bodies also need less collection enhancement over the thermal flux (5.6 times for a 3U at 600 km against 15 times for the compact body), so they rest on less extrapolation, not more. These numbers are estimates from the fitted collection law, and CubeSat radii sit at 25–60 Debye lengths where that law has not been measured; a thin-sheath estimate puts the 3U float near 60 V at 500 km near solar maximum, which would drop that margin below one (Sec. 8).

6 Comparison with micropropulsion

Table 8 compares the device with flown systems in the nanonewton-to-micronewton range. The thrust per watt is about 200 times below an ion thruster, which is Eq. (1) and cannot be engineered away. At micronewtons the power penalty reaches watts and electrospray or FEEP wins. The difference from every flown system is the hardware column: no tank, no feed system, no neutralizer, no total-impulse limit. Nothing in the table produces thrust below 100 nN or fits a gram-scale spacecraft, and that is where the device holds altitude on milliwatts.

Related concepts share the return-circuit idea. Bare electrodynamic tethers [6, 7] and Janhunen’s electric sail and plasma brake [8, 9] all use continuous electron emission with the ambient plasma closing the circuit, but produce thrust by Lorentz force or ion deflection on kilometre-scale tethers. Atmosphere-breathing electric propulsion [10] collects ambient gas but keeps an ionization stage, an acceleration stage and a neutralizer. A

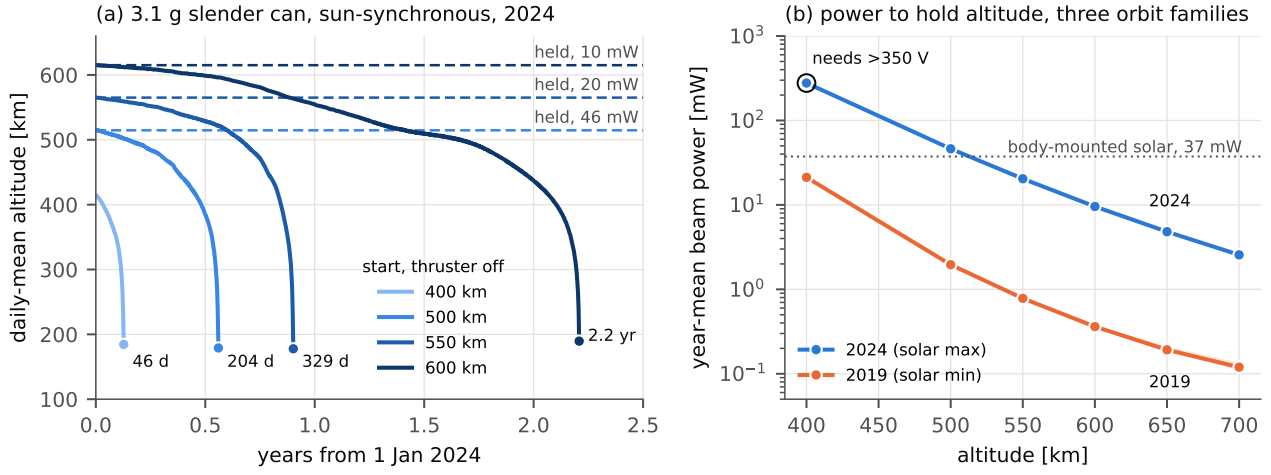


Figure 5: (a) The 3 g slender body on the sun-synchronous orbit in 2024. Solid: thruster off, daily-mean altitude until re-entry. Dashed: the same orbits held by the thruster, with the year-mean beam power; at 400 km the demand exceeds the simulated envelope. (b) Year-mean beam power to hold altitude; line = median of the three orbit families, band = their range. The ring marks 400 km near solar maximum, where the peaks need more than 350 V. A 3U-size body of the same shape needs 100 times these powers.

Table 7: Voltage-following station keeping of the slender body: for each 5-minute point, the supply voltage that cancels that point's drag for the least beam power, with the floating potential solved from the IRI-2020 plasma at that point. Ranges span the three orbit families. Max dip: largest altitude deficit when the same year is replayed with thrust capped at the measured 350 V capability.

altitude	mean P	2024 (solar maximum)				2019
		worst 5 min	$V > 350$ V	φ med / 99th	max dip	
400 km	272–292 mW	1.0–1.5 W	44–47 %	18–50 / 109–276 V	not held	21 mW
500 km	46–48 mW	235–316 mW	0–2.5 %	9–31 / 57–152 V	9 m	2.0 mW
550 km	20–21 mW	120–169 mW	0–1.1 %	7–23 / 41–116 V	4 m	0.8 mW
600 km	9.6–9.9 mW	61–96 mW	0–0.4 %	6–16 / 29–89 V	2 m	0.4 mW
650 km	4.8–5.0 mW	33–53 mW	0–0.2 %	4–11 / 18–69 V	1 m	0.2 mW
700 km	2.5–2.7 mW	18–30 mW	0–0.1 %	3–7 / 14–54 V	1 m	0.1 mW

recent review of propellantless propulsion [11] does not list a standalone ionospheric electron-exchange thruster.

7 Bench experiment

The electrode topology was tested in a laboratory vacuum chamber. A W5W automotive bulb filament with the glass removed, held at -50 and -62 V on its two terminals, faces a grounded 4×4 cm copper plate in a glass chamber at $4\text{--}5$ Pa. The ammeter between plate and earth reads a steady 87 mA once the filament is hot: a single net-negative electrode emits electrons and accelerates them across a gap onto a grounded collector. At this pressure the gap is collisional, so the current includes gas amplification and the result is qualitative. The experiment confirms the principle tested numerically in ladder stage 1: a negative cathode emits and accelerates electrons toward a grounded plate. Controls (bias off, filament cold) and an I–V sweep are the next step.

8 Limitations

Ordered by how much each could change the answer.

- Ion clock and reduced ion mass.** Ions are $400 m_e$. The 3D field campaign settled its floats at $6 \mu\text{s}$ on that clock and found the 800 ns snapshots at about half the settled float, with thrust moving under 2 %; real O^+ is 73 times heavier still and settles roughly $9 \times$ slower, which no run has reached.
- Prescribed beam.** Beam current is a fixed input, not self-consistently emitted from a modelled cathode, and it is launched 0.3 mm above the cathode, which costs 18 % of V in exhaust energy (Sec. 4.4.1); thrust and efficiency are conservative by about 11 % and 19 %. The choice of flight emitter is outside the scope of this paper.
- CubeSat sheath regime.** Runs sit at $r/\lambda_D \approx 2.5$; CubeSats are $25\text{--}60 \lambda_D$ across, where orbital-motion-limited collection no longer applies. A single large-body run would settle this.
- Stationary plasma, held attitude.** No ram flow is modelled, and the mission cases assume a fixed pose without an attitude-control model.
- Orbit model.** Orbits are circular. Free-fall runs starting in 2024 use forecast solar indices after mid-2025; the 2019 runs are observed throughout. The

Table 8: Position against flight micropropulsion in the nanonewton-to-micronewton range.

system	example	thrust	power	I_{sp} [s]	hardware
this work	electron exchange	3–43 nN	12–278 mW	10^6	HV supply, cathode, aperture
cold gas	GomSpace NanoProp [12]	0.1–10 mN	<2 W	60–110	tanks, valves, nozzles
PPT	PPTCUP [13]	40 μ N	2 W	655	capacitor bank, electrodes
electrospray	Busek CMNT [12]	5–30 μ N	1–16 W	200–300	tanks, PPU, neutralizer
FEEP (1U)	NanoFEEP [2]	0–8 μ N	0.2–0.3 W	6000	liquid-metal needles, heater
FEEP	Enpulsion IFM Nano [14]	10–500 μ N	8–40 W	2000–5000	emitter, 2 neutralizers, PPU

altitude-hold replay ignores the denser air a dipping craft meets, which is negligible for the metre-scale dips reported and is flagged where a case does not hold.

All powers are beam power VI ; converter and cathode overheads belong to the flight design.

9 Conclusion

A cathode, a high-voltage supply and the spacecraft’s own skin make a thruster that needs no propellant. PIC simulations measure its operating envelope at 3–43 nN from 12–278 mW, with two constants describing every run to within 3 %, a floating potential fixed by a measured collection law, and a return current that costs 1–9 % of the beam thrust. Against a year of real drag on three orbit families at both ends of the solar cycle, with the float included, a 3 g slender body that re-enters within a year from 550 km without the thruster holds 500–700 km with it for 2.5–48 mW of beam power near solar maximum and 0.1–2 mW near minimum, within its own body-mounted solar supply from 550 km up. The feasibility condition depends on shape and altitude but not on size, so a 3U CubeSat inherits it at about 1 W at 600 km near solar maximum and tens of milliwatts near minimum. The device is cheap to test: it needs no hardware beyond a supply and an electron gun of the kind that already flies as a neutralizer, and it can be retrofitted to an existing ion or Hall thruster after its propellant is spent. The transverse geomagnetic field, once the largest open question, is now measured: at flight strength it leaves the operating point unchanged, and at ten times flight strength it forces field-aligned firing. The flight emitter is the next work.

Data. All simulation inputs, run manifests, gate policies, the orbit propagations and the model are committed in the project repository, archived at DOI:10.5281/zenodo.22115924.

Acknowledgements. This research used the open-source particle-in-cell code WarpX. Primary WarpX contributors are with LBNL, LLNL, CEA-LIDYL, SLAC,

DESY, CERN, Helion Energy, TAE Technologies, and Realta Fusion. We acknowledge all WarpX contributors. Orbit propagation used TudatPy [5].

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